

# ACTL2102 Tutorial

## Week 2: Discrete-Time Markov Chains

Nick Guo

School of Risk and Actuarial Studies, UNSW Sydney

T3 2026

# Plan

- ▶ Main questions: **2.1, 2.2, 2.4 (parts 1–5), 2.10, 2.12**
- ▶ If time permits: 2.9

## Three More Simulation Methods

Each uses a relationship between distributions in place of the inverse c.d.f.

**Normal, by Box–Muller.** If  $U_1, U_2$  are independent  $U(0, 1)$ , then

$$X = (-2 \ln U_1)^{1/2} \cos(2\pi U_2), \quad Y = (-2 \ln U_1)^{1/2} \sin(2\pi U_2)$$

are independent standard normals, and  $\mu + \sigma X$  is  $N(\mu, \sigma^2)$ .

**Gamma( $n, \lambda$ ).** A sum of  $n$  independent  $\text{Exp}(\lambda)$  variables, so

$$X = -\frac{1}{\lambda} \sum_{i=1}^n \ln(U_i).$$

Here  $\lambda$  is a rate, so the mean is  $n/\lambda$ . Some references use the scale  $1/\lambda$  instead.

**Chi-squared.**  $\chi_n^2 = \sum_{i=1}^n Z_i^2$  with  $Z_i \sim N(0, 1)$  independent.

# Stochastic Processes and the Markov Property

A **stochastic process** is a collection of random variables  $\{X(t), t \in T\}$ .  $T$  is the **index set**, usually time, the values lie in the **state space**  $S$ , and one realisation of the whole process is a **sample path**.

## The Markov property

For all  $n \geq 0$  and all states  $i, j, i_0, \dots, i_{n-1}$ ,

$$\begin{aligned}\Pr(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) \\ = \Pr(X_{n+1} = j \mid X_n = i).\end{aligned}$$

Here  $n$  is the present time:  $X_n$  is the current state,  $X_{n+1}$  the next state and  $X_0, \dots, X_{n-1}$  the history. Given the present state, the past states have no influence on the future.

The shorthand  $\Pr(X_{n+1} \mid X_n, \dots, X_0) = \Pr(X_{n+1} \mid X_n)$  means this equality for every choice of states.

# Discrete-Time Markov Chains

A **Markov chain** is a Markov process with a **discrete state space**  $S$ . In Module 2 time is also discrete, a **DTMC**, and  $X_n$  is the state at time  $n$ . In Module 4 time is continuous, a **CTMC**.

The **one-step transition probabilities** are

$$P_{ij}^{(n,n+1)} = \Pr(X_{n+1} = j \mid X_n = i),$$

and if they do not depend on  $n$  the chain is **homogeneous**, written  $P_{ij}$ .

**The transition matrix**

$\mathbf{P} = (P_{ij})$  has  $P_{ij} \geq 0$  and **every row sums to 1**. Row  $i$  is the distribution of the next state given the current state  $i$ , so rows are “from” and columns are “to”.

The lecture notes number the states from 0 and some exercises from 1.

# Chapman–Kolmogorov

For a homogeneous chain the  **$n$ -step transition probability**  $P_{ij}^{(n)} = \Pr(X_{n+m} = j \mid X_m = i)$  does not depend on  $m$ , and

$$P_{ij}^{(n+m)} = \sum_k P_{ik}^{(n)} P_{kj}^{(m)} \quad \text{for all } n, m \geq 0.$$

The sum conditions on the state  $k$  occupied after  $n$  steps. Setting  $m = 1$  gives the forward equations and  $n = 1$  the backward equations.

**The  $n$ -step matrix is  $\mathbf{P}^n$**

The  $(i, j)$  entry of  $\mathbf{P}\mathbf{P}$  is  $\sum_k P_{ik} P_{kj}$ , which is the sum above with  $n = m = 1$ , so  $\mathbf{P}^2$  is the matrix of two-step probabilities. By induction the matrix of  $n$ -step probabilities is  $\mathbf{P}^n$ .

With  $\alpha$  the distribution of  $X_0$ , the unconditional distribution is  $\Pr(X_n = j) = \sum_i \alpha_i P_{ij}^{(n)} = (\alpha \mathbf{P}^n)_j$ .

# Notation

---

|                |                                                               |
|----------------|---------------------------------------------------------------|
| $X_n$          | the state at time $n$ , random; $i, j$ are particular states  |
| $X(t)$         | the general form; in discrete time $X_n = X(t_n)$ , $t_n = n$ |
| $P_{ij}$       | one-step probability from $i$ to $j$ , a number               |
| $\mathbf{P}$   | the matrix of them; bold marks matrices and vectors           |
| $P_{ij}^{(n)}$ | $n$ -step probability, the $(i, j)$ entry of $\mathbf{P}^n$   |
| $\alpha$       | the distribution of $X_0$ , a row vector                      |

---

Two conventions:

- ▶ The full one-step notation  $P_{ij}^{(n, n+1)}$  records the time  $n$  of the step. Writing  $P_{ij}$  assumes homogeneity.
- ▶ The lecture notes write  $P_{ij}^n$ , without brackets, for the  $n$ -step probability. It is the  $(i, j)$  entry of  $\mathbf{P}^n$  and is not the  $n$ th power of the number  $P_{ij}$ . Where a power is meant, bracket the base:  $(P_{ij})^n$ .

## Example: Two-Step Probabilities

Coin 1 lands heads with probability 0.7 and coin 2 with probability 0.6. After heads, coin 1 is flipped the next day; after tails, coin 2. Let  $X_n$  be the coin flipped on day  $n$ :

$$\mathbf{P} = \begin{pmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{pmatrix} \implies \mathbf{P}^2 = \begin{pmatrix} 0.67 & 0.33 \\ 0.66 & 0.34 \end{pmatrix}.$$

For the (1, 1) entry, the coin flipped on the intermediate day is either coin 1 or coin 2:

$$P_{11}^{(2)} = P_{11}P_{11} + P_{12}P_{21} = 0.7(0.7) + 0.3(0.6) = 0.67.$$

**Unconditional distribution.** If each coin is equally likely on day 0, then  $\alpha = (0.5, 0.5)$  and

$$\Pr(X_2 = 1) = 0.5(0.67) + 0.5(0.66) = 0.665.$$

Each row of  $\mathbf{P}^2$  sums to 1.

## Exercise 2.1: Matrix warm-up: products and inverses

**Exercise 2.1:** [dmcpref1, Solution] You are given the following vector and matrices:

$$A = \begin{pmatrix} 3 \\ 1 \\ 10 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 5 & 2 \\ 3 & 12 & 9 \\ 1 & 4 & 7 \end{pmatrix}, \quad C = \begin{pmatrix} 3 & 1 & 2 \\ 2 & 5 & 9 \\ 1 & 3 & 8 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 3 \\ 5 & 7 \end{pmatrix}$$

Calculate the following:

1.  $A'B$
2.  $BC$
3.  $D^{-1}$

# Simulating a Markov Chain

From state  $i$ , the next state is a discrete random variable with probabilities given by **row  $i$  of  $\mathbf{P}$** . It is simulated by the discrete inverse transform of Module 1 applied to that row, using one uniform random number per step.

**Example.** For the two-coin chain, row 1 is (0.7, 0.3) and row 2 is (0.6, 0.4), so the next state is 1 if  $u \leq 0.7$  from state 1, or if  $u \leq 0.6$  from state 2. Starting from  $X_0 = 1$ :

| Step $n$                | 1    | 2    | 3    |
|-------------------------|------|------|------|
| Current state $X_{n-1}$ | 1    | 1    | 2    |
| $u_n$                   | 0.43 | 0.85 | 0.21 |
| Next state $X_n$        | 1    | 2    | 1    |

The row used at each step is that of the current state.

# Enlarging the State Space

If the next value depends on the last *two* values,  $\{X_n\}$  is not a Markov chain, because the present value alone does not determine the distribution of the next.

A Markov chain is recovered by letting the state record both values:

$$Y_n = (X_{n-1}, X_n).$$

If  $X_n$  takes  $N$  values then  $Y_n$  takes up to  $N^2$ , and  $\mathbf{P}$  is written for these pairs.

## Working with $Y_n$

From  $(a, b)$  the only possible next states are of the form  $(b, c)$ , so most entries of  $\mathbf{P}$  are zero. A question about  $X_n$  is answered by adding the probabilities of the states of  $Y_n$  that make up the event.

If the dependence extends back  $k$  steps, the states are  $k$ -tuples.

## Exercise 2.2: Two-day weather chain

**Exercise 2.2:** [dmc1, Solution] Ross: Example 4.9, Suppose that whether or not it rains today depends on previous weather conditions through the last two days. Specifically, suppose that if it has rained for the past two days, then it will rain tomorrow with probability 0.7; if it rained today but not yesterday, then it will rain tomorrow with probability 0.5; if it rained yesterday but not today, then it will rain tomorrow with probability 0.4; if it has not rained in the past two days, then it will rain tomorrow with probability 0.2.

If we know that it rained on both Monday and Tuesday, what is the probability that it will rain on Thursday.

## Exercise 2.10: Claims process: designing a Markov state space

**Exercise 2.10:** [dmc9, Solution] An insurance company can receive either 1, 2, or 3 claims per month. The probabilities of the number of claims per month depend on the state of the world. If the last state of the world was:

- “many claims”: the insurer will receive next period either 2 or 3 claims with equal probability;
- “few claims”: the insurer will receive next period either 1 or 2 claims with equal probability.

The current state of world depend on the number of claims:

- If there is 1 claim, the state of the world will be “few claims”,
- if there are 3 claims, the state of the world will be “many claims”;
- if there are 2 claims, the both states of the world occur with equal probability.

1. Define the time index of the Markov process.
2. Define the state space for the Markov process of the aggregate number of claims.

**Hint:** The state space should be defined such that the process has the Markov property.

# Accessibility, Communication, Classes

- ▶ State  $i$  is **absorbing** if  $P_{ii} = 1$ .
- ▶  $j$  is **accessible** from  $i$ , written  $i \rightarrow j$ , if  $P_{ij}^{(n)} > 0$  for some  $n \geq 0$ . Since  $\mathbf{P}^0 = \mathbf{I}$ ,  $i \rightarrow i$  always holds.
- ▶  $i$  and  $j$  **communicate**, written  $i \leftrightarrow j$ , if  $i \rightarrow j$  and  $j \rightarrow i$ .
- ▶ The **class** of  $i$  is  $C(i) = \{j \in S : i \leftrightarrow j\}$ .

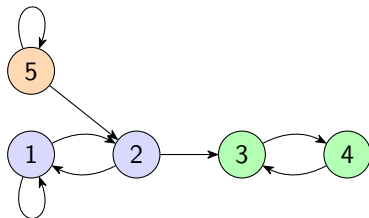
Communication is an equivalence relation, so the classes partition  $S$ : two classes are either identical or disjoint. A chain is **irreducible** if there is only one class.

$P_{ij} = 0$  does not imply that  $j$  is inaccessible from  $i$ , since a path  $i \rightarrow k \rightarrow j$  through another state may exist.

**Method.** Draw the transition diagram, list the states accessible from each state, and group the states that communicate.

## Example: Communicating Classes

The classes depend only on which transitions are possible.



States 1 and 2 each lead to  $\{1, 2, 3, 4\}$ , states 3 and 4 only to  $\{3, 4\}$ , and state 5 to every state. The classes are  $\{1, 2\}$ ,  $\{3, 4\}$  and  $\{5\}$ , so the chain is not irreducible.

### Closed and open classes

A class is **closed** if no state outside it is accessible from it, and **open** otherwise. Here only  $\{3, 4\}$  is closed. A finite closed class is recurrent and an open class is transient (Week 3).

## Exercise 2.4: Absorbing states, accessibility and classes (parts 1–5)

**Exercise 2.4:** [dmc3, Solution] For the following transition matrix with states  $i = 1, \dots, 7$ :

$$\begin{bmatrix} 0 & 0.8 & 0.2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 & 0 & 0 & 0 \\ 0.6 & 0 & 0 & 0 & 0 & 0.4 & 0 \\ 0 & 0 & 0.6 & 0 & 0.4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

1. Identify the absorbing states;
2. For states  $i = 1, \dots, 7$ , identify which states are accessible;
3. Identify the communicating classes;
4. Is the Markov chain irreducible?
5. Which states are recurrent and which are transient?

## Exercise 2.12: Classify the states of four chains

**Exercise 2.12:** [dmc11, Solution] Ross: Chapter 4, Q14. For each the following Markov chain:

$$\mathbf{P}_1 = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$$

$$\mathbf{P}_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$\mathbf{P}_3 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$\mathbf{P}_4 = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & \frac{2}{3} & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

1. Specify the classes of the the Markov chains;
2. Determine for all states whether they are transient or recurrent.

# Common Mistakes

| Not this                                | This                                                              |
|-----------------------------------------|-------------------------------------------------------------------|
| $P_{ij} = 0$ , so $i \not\rightarrow j$ | $i \rightarrow j$ is possible via $i \rightarrow k \rightarrow j$ |
| $\Pr(X_n = j) = P_{ij}^{(n)}$           | $\Pr(X_n = j) = (\alpha \mathbf{P}^n)_j$                          |
| $\mathbf{P}^n \alpha$                   | $\alpha \mathbf{P}^n$ , row vector on the left                    |
| $(P_{ij})^n$                            | $P_{ij}^{(n)}$ , the $(i, j)$ entry of $\mathbf{P}^n$             |
| an answer left in pairs $(a, b)$        | the pairs summed, in terms of $X_n$                               |

Every row of  $\mathbf{P}$  and of  $\mathbf{P}^n$  sums to 1.

## Exercise 2.9: Weather depending on the last three days

**Exercise 2.9:** [dmc8, Solution] Ross: Chapter 4, Q2. Suppose that whether or not it rains today depends on previous weather conditions through the last three days. Show how this system may be analysed by using a Markov chain. How many states are needed? Define the states.

Thanks for coming!